Photon-mediated correlated hopping in a synthetic ladder: Supplemental Materials

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S1. THEORY MODEL

In this section we describe how to obtain Eq. (2) in the main text beginning with the fundamental interaction between atoms and light in a cavity. We work with ground and excited state manifolds of N atoms, which are assumed to have hyperfine spin F_g and F_e, respectively and are split by an energy ω_a. The ground and excited states of atom i are defined with respect to a quantization axis that is parallel to the cavity axis, and we label them as $|g_m\rangle_i$ and $|e_m\rangle_i$, respectively, where m is the magnetic quantum number. These atoms interact with two circularly polarized cavity modes at frequency ω_c, with strength g_i for atom i. These coupling constants are parameterized as $g_i = g_c\eta_i$, where η_i are dimensionless numbers of order 1 that describe the inhomogeneity in the couplings. The Hamiltonian describing this interaction is then [S1]

$$\hat{H}_{\text{atom-light}}/\hbar = \omega_a\hat{N}_e + \omega_c(\hat{a}_L^\dagger\hat{a}_L + \hat{a}_R^\dagger\hat{a}_R) + g_c(\hat{a}_L\hat{L}^+ + \hat{a}_R\hat{R}^+ + h.c.), \tag{S1}$$

where $\hat{a}_L(\hat{a}_R)$ are boson operators that annihilate a σ^- (σ^+) photon, $\hat{N}_e = \sum_{i,m}|e_m\rangle\langle e_m|_i$ is the number of atoms in the excited state, $\hat{L}^+ = \sum_{i,m}\eta_i C_m^{-1}|e_{m-1}\rangle\langle g_m|_i$ and $\hat{R}^+ = \sum_{i,m}\eta_i C_m^{+1}|e_{m+1}\rangle\langle g_m|_i$ are multilevel collective dipole operators with σ^- and σ^+ polarization, and $C_m^p = \langle F_g, m; 1, p|F_e, m+p\rangle$ are Clebsch-Gordan coefficients. The interaction describes the absorption of a σ^- (σ^+) cavity photon by the atomic ensemble accompanied by an atomic excitation with σ^- (σ^+) light, together with the inverse process.

We consider the situation in which the cavity is far detuned from the atomic transition $|\Delta_c| \gg g_c\sqrt{N}, \kappa$, where $\Delta_c = \omega_c - \omega_a$, and κ is the cavity decay rate. In addition to this, the cavity will be driven by two tones with frequencies close to ω_a, so they are far away from the cavity resonance frequency. Because of this, the number of intracavity photons is always very small, hence the light degrees of freedom can be adiabatically eliminated. The atom-light interaction, albeit weak, induces many-body splittings of the excited states, which the drives are designed to probe. The cavity-mediated atom-atom interactions are described by the following effective Hamiltonian [S1]

$$\hat{H}_{\text{atom-atom}}/\hbar = \omega_a\hat{N}_e + \chi\hat{L}^+\hat{L}^- + \chi\hat{R}^+\hat{R}^-, \tag{S2}$$

where $\hat{L}^- = (L^+)^\dagger$, $\hat{R}^- = (R^+)^\dagger$ and $\chi = -g_c^2/\Delta_c$ is the effective atom-atom interaction. The two injected driving fields (A, B) are assumed to be σ^- polarized, with frequencies $\omega_d^{A,B}$ and intensities outside the cavity of $|\alpha_{A,B}|^2$ photons per second. These injected fields establish a small intracavity field that induces Rabi flopping on the atoms with Rabi frequencies $\Omega_{A,B} = -\sqrt{\kappa}\alpha_{A,B}g_c/\Delta_c$. Adding these terms leads to Eq. (1) in the main text,

$$\hat{H}_{\text{aa+d}}/\hbar = \omega_a\hat{N}_e + \chi\hat{L}^+\hat{L}^- + \chi\hat{R}^+\hat{R}^- + \Big[\Big(\Omega_A e^{-i\omega_d^A t} + \Omega_B e^{-i\omega_d^B t}\Big)\hat{L}^\dagger + h.c.\Big]. \tag{S3}$$

When the intracavity Rabi frequencies are small compare to $\max(\chi N, |\Delta_{A,B}|)$, with $\Delta_{A,B} = \omega_d^{A,B} - \omega_a$, the excited state will only be virtually populated and can thus be adiabatically eliminated. To make the discussion simple, we consider first the case of a single drive Ω_A, with associated Hamiltonian

$$\hat{H}_A/\hbar = \omega_a\hat{N}_e + \chi\hat{L}^+\hat{L}^- + \chi\hat{R}^+\hat{R}^- + \Big(\Omega_A e^{-i\omega_d^A t}\hat{L}^+ + \Omega_A^* e^{i\omega_d^A t}\hat{L}^-\Big). \tag{S4}$$

In the rotating frame of the drive, this becomes

$$\hat{H}'_A/\hbar = \underbrace{-\Delta_A\hat{N}_e + \chi\hat{L}^+\hat{L}^- + \chi\hat{R}^+\hat{R}^-}_{\hat{H}_0} + \underbrace{\Big(\Omega_A\hat{L}^+ + \Omega_A^*\hat{L}^-\Big)}_{\text{perturbation}}. \tag{S5}$$

Given that the Hamiltonian in this rotating frame is time-independent, the ground state effective description can be calculated using perturbation theory. First, notice that all the ground states are degenerate, with energy 0, in the absence of the perturbation. Furthermore, since the operator $\hat{L}^+$ maps the ground state manifold to single excitation

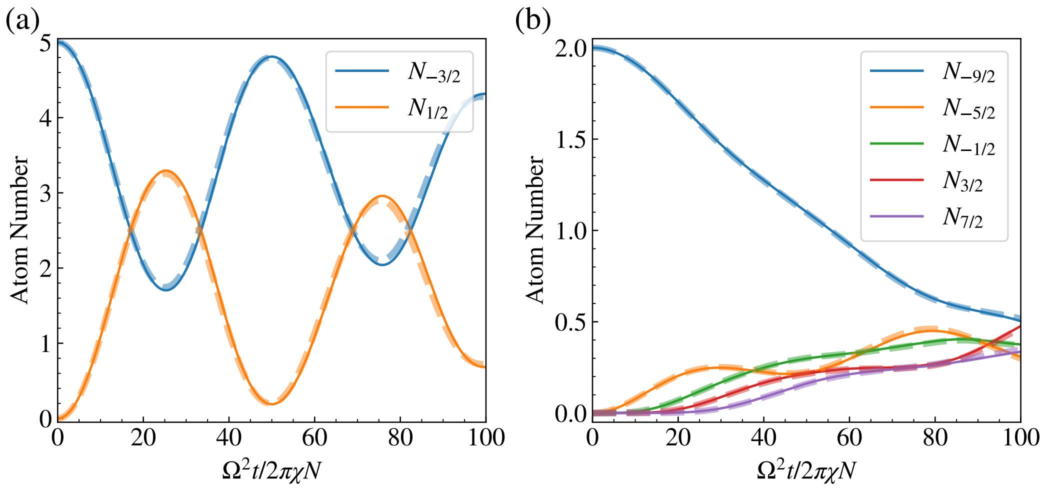

FIG. S1. Numerical benchmarking of adiabatic elimination of atomic excited states. The Hamiltonian dynamics under Eq. (S3) are shown in solid lines, while the Hamiltonian dynamics under Eq. (S16) are shown in dashed lines. We set $\Delta_A = -3\chi N$, $\Delta_B = 4.1\chi N$, $\Omega_{A,B} = \Omega = 0.05\chi N$. We consider the following two cases: (a) 4-level system with initial state $|g_{-3/2}\rangle^{\otimes(N/2)}|g_{3/2}\rangle^{\otimes(N/2)}$, and atom number $N = 10$. (b) 10-level system with initial state $|g_{-9/2}\rangle^{\otimes(N/2)}|g_{9/2}\rangle^{\otimes(N/2)}$, and atom number $N = 4$.

states, the first order perturbative correction due to the drive is 0. The degeneracy of the ground states is lifted at second order using degenerate perturbation theory,

$$\hat{H}_{\text{eff},A}/\hbar \approx -\hat{P}_g\left(\Omega_A\hat{L}^+ + \Omega_A^*\hat{L}^-\right)\frac{1}{\hat{H}_0}\left(\Omega_A\hat{L}^+ + \Omega_A^*\hat{L}^-\right)\hat{P}_g = -|\Omega_A|^2\hat{P}_g\hat{L}^-\frac{1}{\hat{H}_0}\hat{L}^+\hat{P}_g, \tag{S6}$$

where $\hat{P}_g$ are projectors onto the ground state manifold. The unperturbed Hamiltonian $\hat{H}_0$ includes not only the single particle term $\propto \hat{N}_e$ but also nonlinear contributions $\propto \hat{L}^+\hat{L}^-$, $\hat{R}^+\hat{R}^-$ and thus obtaining a simple expression for $\hat{H}_0^{-1}$ is not straightforward. Fortunately, $\hat{H}_0$ conserves the number of excitations, which is a strong enough symmetry to allow for an explicit calculation of $\hat{H}_0^{-1}$. To do this, consider the quantity

$$\hat{H}_0\hat{L}^+\hat{P}_g = -\Delta_A\hat{N}_e\hat{L}^+\hat{P}_g + \chi\hat{L}^+\hat{L}^-\hat{L}^+\hat{P}_g + \chi\hat{R}^+\hat{R}^-\hat{L}^+\hat{P}_g. \tag{S7}$$

Due to the projectors, some parts of this expression can be simplified. First, $\hat{N}_e\hat{L}^+\hat{P}_g = \hat{L}^+\hat{P}_g$ since the image of the operator $\hat{L}^+\hat{P}_g$ is on the single excitation manifold. Second, the operators $\hat{L}^-\hat{L}^+\hat{P}_g$ and $\hat{R}^-\hat{L}^+\hat{P}_g$ preserve the ground state manifold and can thus be written as $\hat{L}^-\hat{L}^+\hat{P}_g \equiv \hat{D}_L\hat{P}_g$ and $\hat{R}^-\hat{L}^+\hat{P}_g \equiv \hat{T}^-\hat{P}_g$ where

$$\begin{aligned} \hat{D}_L &\equiv \hat{P}_g\hat{L}^-\hat{L}^+\hat{P}_g = \sum_i \hat{D}_i^L, \quad \hat{D}_i^L = \sum_m \eta_i^2 (C_m^{-1})^2 |g_m\rangle\langle g_m|_i, \\ \hat{T}^- &\equiv \hat{P}_g\hat{R}^-\hat{L}^+\hat{P}_g = \sum_i \hat{T}_i^-, \quad \hat{T}_i^- = \sum_m \eta_i^2 C_{m-2}^{+1} C_m^{-1} |g_{m-2}\rangle\langle g_m|_i, \end{aligned} \tag{S8}$$

are collective ground state operators only. Therefore

$$\boxed{\hat{H}_0\hat{L}^+\hat{P}_g = (\hat{L}^+\hat{P}_g)(-\Delta_A + \chi\hat{D}_L) + (\hat{R}^+\hat{P}_g)(\chi\hat{T}^-)}. \tag{S9}$$

Note the ordering of the operators, which is important since $[\hat{L}^\pm, \hat{D}_L] \neq 0$ and $[\hat{R}^\pm, \hat{T}^-] \neq 0$. Similarly,

$$\boxed{\hat{H}_0\hat{R}^+\hat{P}_g = (\hat{R}^+\hat{P}_g)(-\Delta_A + \chi\hat{D}_R) + (\hat{L}^+\hat{P}_g)(\chi\hat{T}^+)}, \tag{S10}$$

where

$$\begin{aligned} \hat{D}_R &\equiv \hat{P}_g\hat{R}^-\hat{R}^+\hat{P}_g = \sum_i \hat{D}_i^R, \quad \hat{D}_i^R = \sum_m \eta_i^2 (C_m^{+1})^2 |g_m\rangle\langle g_m|_i, \\ \hat{T}^+ &\equiv \hat{P}_g\hat{L}^-\hat{R}^+\hat{P}_g = \sum_i \hat{T}_i^+, \quad \hat{T}_i^+ = \sum_m \eta_i^2 C_m^{+1} C_{m+2}^{-1} |g_{m+2}\rangle\langle g_m|_i. \end{aligned} \tag{S11}$$

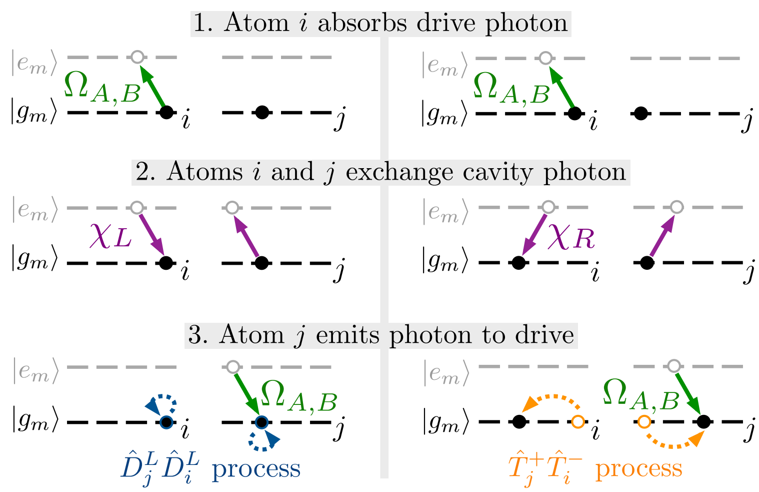

FIG. S2. Perturbative ground-ground processes between atom i and j. (a) The $\hat{D}_L\hat{D}_L$ process exchanging left circular polarized photons. (b) The $\hat{T}^+\hat{T}^-$ process exchanging right circular polarized photons. Here we define $\chi_L = \chi_R = \chi$, with subscript L or R labelling photon polarizations.

Right multiplying Eq. (S10) by $(-\Delta_A + \chi\hat{D}_R)^{-1}\chi\hat{T}^-$ and subtracting the result from Eq. (S9) results in

$$\hat{H}_0\Big[\hat{L}^+\hat{P}_g - \hat{R}^+\hat{P}_g\big(-\Delta_A + \chi\hat{D}_R\big)^{-1}\chi\hat{T}^-\Big] = (\hat{L}^+\hat{P}_g)\Big[\big(-\Delta_A + \chi\hat{D}_L\big) - \chi^2\hat{T}^+\big(-\Delta_A + \chi\hat{D}_R\big)^{-1}\hat{T}^-\Big]. \tag{S12}$$

Left multiplying by $\hat{P}_g\hat{L}^-\hat{H}_0^{-1}$ leads to

$$\hat{D}_L - \hat{T}^+\big(-\Delta_A + \chi\hat{D}_R\big)^{-1}\chi\hat{T}^- = (\hat{P}_g\hat{L}^-\hat{H}_0^{-1}\hat{L}^+\hat{P}_g)\Big[\big(-\Delta_A + \chi\hat{D}_L\big) - \chi^2\hat{T}^+\big(-\Delta_A + \chi\hat{D}_R\big)^{-1}\hat{T}^-\Big]. \tag{S13}$$

Therefore

$$(\hat{P}_g\hat{L}^-\hat{H}_0^{-1}\hat{L}^+\hat{P}_g) = \frac{1}{\chi} - \frac{\Delta_A}{\chi}\left(\Delta_A - \chi\hat{D}_L - \chi^2\hat{T}^+\hat{G}_R^A\hat{T}^-\right)^{-1}, \tag{S14}$$

where $\hat{G}_R^\nu = (\Delta_\nu - \chi\hat{D}_R)^{-1}$ and the effective ground state Hamiltonian due to drive A is

$$\boxed{\hat{H}_{\text{eff},A}/\hbar = -\frac{|\Omega_A|^2}{\chi} + \frac{|\Omega_A|^2\Delta_A}{\chi}\left(\Delta_A - \chi\hat{D}_L - \chi^2\hat{T}^+\hat{G}_R^A\hat{T}^-\right)^{-1}}. \tag{S15}$$

Similarly, drive B will generate an analogous contribution to the effective ground state Hamiltonian. Since there are no possible interference pathways between the two drives (at second order) due to their different frequencies (we will take $\Delta_A\Delta_B < 0$), the full effective Hamiltonian will be the sum of both contributions

$$\boxed{\hat{H}_{\text{eff}}/\hbar = -\sum_{\nu=A,B}\frac{|\Omega_\nu|^2}{\chi} + \sum_{\nu=A,B}\frac{|\Omega_\nu|^2\Delta_\nu}{\chi}\left(\Delta_\nu - \chi\hat{D}_L - \chi^2\hat{T}^+\hat{G}_R^\nu\hat{T}^-\right)^{-1}}. \tag{S16}$$

Omitting the c-number terms leads to Eq. (2) in the main text. We have numerically benchmarked the validity of $\hat{H}_{\text{eff}}$ [Eq. (S16)] to capture the dynamics of the full atomic Hamiltonian $\hat{H}_{\text{aa+d}}$ [Eq. (S3)]. Fig. S1 shows the excellent agreement between them.

Now we proceed to discuss the reason why we need two external drives. In the off-resonant regime ($|\Delta_{A,B}| \gtrsim |\chi N|$), one can expand $\hat{H}_{\text{eff}}$ in a power series of $\chi N/\Delta_{A,B}$, and keep only the leading order terms as an excellent approximation. It's worth to mention that we only use this approximation for a qualitative understanding of the dynamics in our system, while all the quantitative calculations are still based on Eq. (S16). In this approximation, the effective Hamiltonian simplifies to

$$\hat{H}_{\text{eff}}/\hbar \approx \sum_{\nu=A,B}\frac{|\Omega_\nu|^2}{\Delta_\nu}\hat{D}_L + \frac{|\Omega_\nu|^2\chi}{\Delta_\nu^2}(\hat{D}_L\hat{D}_L + \hat{T}^+\hat{T}^-). \tag{S17}$$

As we discussed in the main text, the first term describes the single-particle AC Stark shift, the second term describes the population dependent collective shift, and the third term describes correlated hopping. The microscopic mechanism of the second and the third term is shown in Fig. S2. If we have only one external drive, the single-particle AC Stark shift term always play dominant role. In order to observe correlated hopping, one can use two external drives with detunings in opposite sign to suppress the undesirable energy shifts generated by the first two terms. Considering an initial state $|i\rangle$ and a final state $|f\rangle$ connected by correlated hopping processes, $\langle f|\hat{T}^+\hat{T}^-|i\rangle \neq 0$, the condition of suppressing the first two terms can be written as

$$\left|\langle i|\hat{H}_{\text{eff}}|i\rangle - \langle f|\hat{H}_{\text{eff}}|f\rangle\right| \ll \left|\sum_{\nu=A,B}\frac{\hbar|\Omega_\nu|^2\chi}{\Delta_\nu^2}\langle f|\hat{T}^+\hat{T}^-|i\rangle\right|. \tag{S18}$$

This condition can be satisfied by tuning $\Delta_{A,B}$. Even though it is only computed using the initial state and thus only guarantee to be satisfied at $t=0$, we find that in a large parameter regime it allows to make the correlated hopping process to be dominant during a time window where they can give rise to non-trivial spreading through the synthetic ladder.

S2. SCHWINGER BOSONS AND UNDEPLETED PUMP APPROXIMATION

In the main text, we focused on the initial states as a direct product of a permutationally symmetric states of $N/2$ atoms on the upper leg of the synthetic ladder ($|\psi_{\text{up}}\rangle$) and a permutationally symmetric state of $N/2$ atoms on the lower leg ($|\psi_{\text{down}}\rangle$),

$$|\psi\rangle = |\psi_{\text{up}}\rangle \otimes |\psi_{\text{down}}\rangle. \tag{S19}$$

Since the effective ground state Hamiltonian [Eq. (S16)] forbids any direct hopping process between the legs of the synthetic ladder, the Hamiltonian dynamics with these initial states occurs in a reduced Hilbert space consisted of any $SU\left(\frac{2F_g+1}{2}\right) \times SU\left(\frac{2F_g+1}{2}\right)$ rotations on $|\psi\rangle$, where $SU(n)$ denotes special unitary group of $n\times n$ matrices. So we can assign one set of Schwinger bosons $\{\hat{a}_{-F_g+1}, \hat{a}_{-F_g+3}, \cdots, \hat{a}_{F_g}\}$ to the upper leg and another set of Schwinger bosons $\{\hat{a}_{-F_g}, \hat{a}_{-F_g+2}, \cdots, \hat{a}_{F_g-1}\}$ to the lower leg. In this way, we can rewrite the operators acting only on the ground manifold as,

$$\hat{D}_L = \sum_m (C_m^{-1})^2 \hat{a}_m^\dagger \hat{a}_m, \quad \hat{D}_R = \sum_m (C_m^{+1})^2 \hat{a}_m^\dagger \hat{a}_m, \quad \hat{T}^+ = \sum_m C_m^{+1} C_{m+2}^{-1} \hat{a}_{m+2}^\dagger \hat{a}_m, \tag{S20}$$

where $\hat{a}_m$ is the bosonic annihilation operator for state $|g_m\rangle$. In the following, we will apply undepleted pump approximation (UPA) to the four-level and the six-level system discussed in the main text. The cascaded correlated hopping process in the ten-level system is beyond the reach of UPA.

A. UPA for four-level system

We first consider the initial state $|g_{-3/2}\rangle^{\otimes(N/2)}|g_{3/2}\rangle^{\otimes(N/2)}$ for the four-level ($F_g = F_e = 3/2$) system discussed in the main text. The short time dynamics in this system can be understood via UPA, where to the leading order one can replace the bosonic operators of the initial macroscopically occupied states as c-numbers,

$$\hat{a}_{\pm 3/2} = \hat{a}_{\pm 3/2}^\dagger \approx \sqrt{\frac{N}{2}}, \quad \text{and} \quad \hat{a}_{\pm 3/2}^\dagger \hat{a}_{\pm 3/2} = \frac{N}{2} - \hat{a}_{\mp 1/2}^\dagger \hat{a}_{\mp 1/2}. \tag{S21}$$

So the operators in Eq. (S20) can be approximated as

$$\hat{D}_L \approx \frac{N}{5} + \frac{8}{15}\hat{a}_{1/2}^\dagger \hat{a}_{1/2}, \quad \hat{D}_R \approx \frac{N}{5} + \frac{8}{15}\hat{a}_{-1/2}^\dagger \hat{a}_{-1/2}, \quad \hat{T}^+ \approx -\frac{2\sqrt{6N}}{15}(\hat{a}_{1/2}^\dagger + \hat{a}_{-1/2}). \tag{S22}$$

Plugging in this approximation in the effective ground state Hamiltonian [Eq. (S16)], and keeping the terms with operators $\hat{a}_{\pm 1/2}, \hat{a}^\dagger_{\pm 1/2}$ up to second order, we obtain,

$$\begin{aligned}
\hat{H}_{\text{eff}}/\hbar &= \sum_{\nu=A,B} \frac{|\Omega_\nu|^2 \Delta_\nu}{\chi} \left[\Delta_\nu - \chi \hat{D}_L - \chi^2 \hat{T}^+ (\Delta_\nu - \chi \hat{D}_R)^{-1} \hat{T}^- \right]^{-1} \\
&\approx \sum_{\nu=A,B} \frac{|\Omega_\nu|^2 \Delta_\nu}{\chi} \left[(\Delta_\nu - \chi N/5) - \frac{8\chi}{15} \hat{a}^\dagger_{1/2} \hat{a}_{1/2} - \frac{\chi^2}{\Delta_\nu - \chi N/5} \frac{8N}{75} (\hat{a}^\dagger_{1/2} + \hat{a}_{-1/2})(\hat{a}_{1/2} + \hat{a}^\dagger_{-1/2}) \right]^{-1} \\
&\approx K_1 \hat{a}^\dagger_{-1/2} \hat{a}_{-1/2} + K_2 \hat{a}^\dagger_{1/2} \hat{a}_{1/2} + K_3 (\hat{a}^\dagger_{-1/2} \hat{a}^\dagger_{1/2} + \hat{a}_{-1/2} \hat{a}_{1/2}).
\end{aligned} \tag{S23}$$

where

$$\begin{aligned}
K_1 = K_3 &= \sum_{\nu=A,B} \frac{8}{75} \frac{|\Omega_\nu|^2 \Delta_\nu \chi N}{(\Delta_\nu - \chi N/5)^3}, \\
K_2 &= \sum_{\nu=A,B} \frac{8}{15} \frac{|\Omega_\nu|^2 \Delta_\nu}{(\Delta_\nu - \chi N/5)^2} + \frac{8}{75} \frac{|\Omega_\nu|^2 \Delta_\nu \chi N}{(\Delta_\nu - \chi N/5)^3}.
\end{aligned} \tag{S24}$$

The effective Hamiltonian under the UPA [Eq. (S23)] can be exactly diagonalized using a Bogoliubov transformation. In this case the dynamics of the populations in the $|g_{\pm 1/2}\rangle$ states, labelled as $N_{\pm 1/2}$, is found to be given by the following analytic formula in different parameter regimes:

$$N_{\pm 1/2}(t) = \begin{cases} \dfrac{4K_3^2}{(K_1+K_2)^2 - 4K_3^2} \sin^2 \left(\sqrt{(K_1+K_2)^2 - 4K_3^2}\, \dfrac{t}{2} \right) & \text{if } (K_1+K_2)^2 > 4K_3^2 \\ K_3^2 t^2 & \text{if } (K_1+K_2)^2 = 4K_3^2 \\ \dfrac{4K_3^2}{4K_3^2 - (K_1+K_2)^2} \sinh^2 \left(\sqrt{4K_3^2 - (K_1+K_2)^2}\, \dfrac{t}{2} \right) & \text{if } (K_1+K_2)^2 < 4K_3^2 \end{cases}, \tag{S25}$$

which are shown using dashed lines in Fig. 2(c) in the main text. Notice that $N_{\pm 1/2}(t)$ shows small sinusoidal oscillations in the regime $(K_1+K_2)^2 > 4K_3^2$, while it shows an exponential growth in the regime $(K_1+K_2)^2 < 4K_3^2$. This indicates a dynamical phase transition (DPT) at the critical point $(K_1+K_2)^2 = 4K_3^2$, which is equivalent to

$$\sum_{\nu=A,B} \frac{|\Omega_\nu|^2 \Delta_\nu}{(\Delta_\nu - \chi N/5)^2} = 0 \quad \text{or} \quad \sum_{\nu=A,B} \frac{|\Omega_\nu|^2 \Delta_\nu}{(\Delta_\nu - \chi N/5)^2} + \frac{4}{5} \frac{|\Omega_\nu|^2 \Delta_\nu \chi N}{(\Delta_\nu - \chi N/5)^3} = 0. \tag{S26}$$

By solving these two equations, one can obtain the phase boundaries shown in Fig. 2(b) in the main text.

B. UPA for six-level system

In the main text, we consider the initial state $[\sqrt{p_{-3/2}}|g_{-3/2}\rangle + \sqrt{p_{5/2}}|g_{5/2}\rangle]^{\otimes(N/2)}|g_{-1/2}\rangle^{\otimes(N/2)}$ for the six-level ($F_g = F_e = 5/2$) system, where $p_{5/2} = 1 - p_{-3/2}$. Similar to the previous subsection, we use the UPA to analyze the short time dynamics in this system. We have 6 independent Schwinger bosons in this problem,

$$\begin{aligned}
&\text{Empty}: \quad \hat{a}_{-5/2}, \quad \hat{a}_{3/2}, \quad \hat{a}_{1/2}, \quad \hat{a}_- = -\sqrt{p_{5/2}}\hat{a}_{-3/2} + \sqrt{p_{-3/2}}\hat{a}_{5/2} \\
&\text{Occupied}: \quad \hat{a}_{-1/2}, \quad \hat{a}_+ = \sqrt{p_{-3/2}}\hat{a}_{-3/2} + \sqrt{p_{5/2}}\hat{a}_{5/2}
\end{aligned}$$

We would like to replace the occupied modes by c-numbers, and keep the empty modes up to second order. So we have

$$\begin{aligned}
\hat{D}_L &= \frac{16}{35}\hat{a}^\dagger_{-1/2}\hat{a}_{-1/2} + \frac{16}{35}\hat{a}^\dagger_{3/2}\hat{a}_{3/2} + \frac{18}{35}\hat{a}^\dagger_{1/2}\hat{a}_{1/2} + \frac{2}{7}\hat{a}^\dagger_+\hat{a}_+ + \frac{2}{7}\hat{a}^\dagger_-\hat{a}_- \\
&\approx \frac{13}{35}N - \frac{16}{35}\hat{a}^\dagger_{-5/2}\hat{a}_{-5/2} + \frac{8}{35}\hat{a}^\dagger_{1/2}\hat{a}_{1/2},
\end{aligned} \tag{S27}$$

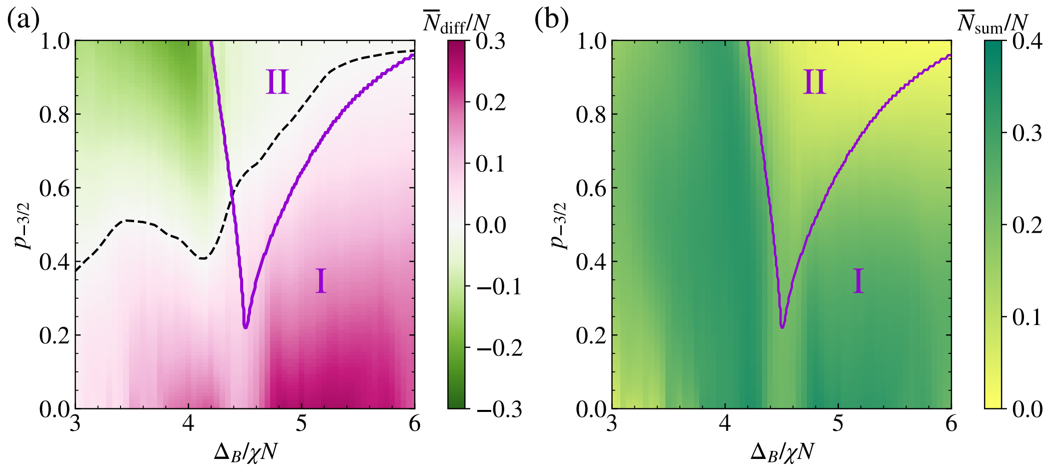

FIG. S3. Dynamical phase boundary in the six-level system (purple line) calculated by UPA in comparison with (a) $\overline{N}_{\rm diff}/N$ and (b) $\overline{N}_{\rm sum}/N$ calculated by exact diagonalization with 20 atoms. We set $\Delta_A = -3\chi N$, and the same initial state for the six-level system as discussed in the main text. The black dashed line in (a) indicates balanced transport.

$$\begin{aligned}\hat{D}_R &= \frac{2}{7}\hat{a}^\dagger_{-5/2}\hat{a}_{-5/2} + \frac{2}{7}\hat{a}^\dagger_{3/2}\hat{a}_{3/2} + \frac{18}{35}\hat{a}^\dagger_{-1/2}\hat{a}_{-1/2} + \frac{16}{35}\hat{a}^\dagger_{1/2}\hat{a}_{1/2} + \frac{16}{35}(\sqrt{p_{-3/2}}\hat{a}^\dagger_+ - \sqrt{p_{5/2}}\hat{a}^\dagger_-)(\sqrt{p_{-3/2}}\hat{a}_+ - \sqrt{p_{5/2}}\hat{a}_-)\\ &\approx \left(\frac{9}{35} + \frac{8}{35}p_{-3/2}\right)N - \frac{8}{35}\hat{a}^\dagger_{-5/2}\hat{a}_{-5/2} - \frac{8}{35}\hat{a}^\dagger_{3/2}\hat{a}_{3/2} + \frac{16}{35}(1-p_{-3/2})\hat{a}^\dagger_{1/2}\hat{a}_{1/2} + \frac{16}{35}(p_{5/2}-p_{-3/2})\hat{a}^\dagger_-\hat{a}_-\\ &\quad - \frac{16}{35}\sqrt{p_{-3/2}p_{5/2}}\sqrt{\frac{N}{2}}(\hat{a}^\dagger_- + \hat{a}_-),\end{aligned} \tag{S28}$$

$$\begin{aligned}\hat{T}^+ &= -\left[\frac{4\sqrt{10}}{35}\hat{a}^\dagger_{-1/2}\hat{a}_{-5/2} + \frac{12\sqrt{2}}{35}\hat{a}^\dagger_{3/2}\hat{a}_{-1/2} + \frac{12\sqrt{2}}{35}\hat{a}^\dagger_{1/2}(\sqrt{p_{-3/2}}\hat{a}_+ - \sqrt{p_{5/2}}\hat{a}_-) + \frac{4\sqrt{10}}{35}(\sqrt{p_{5/2}}\hat{a}^\dagger_+ + \sqrt{p_{-3/2}}\hat{a}^\dagger_-)\hat{a}_{1/2}\right]\\ &\approx -\left[\frac{4\sqrt{10}}{35}\sqrt{\frac{N}{2}}\hat{a}_{-5/2} + \frac{12\sqrt{2}}{35}\sqrt{\frac{N}{2}}\hat{a}^\dagger_{3/2} + \frac{12\sqrt{2}}{35}\sqrt{\frac{p_{-3/2}N}{2}}\hat{a}^\dagger_{1/2} + \frac{4\sqrt{10}}{35}\sqrt{\frac{p_{5/2}N}{2}}\hat{a}_{1/2}\right].\end{aligned} \tag{S29}$$

In this way, the effective ground state Hamiltonian [see Eq. (S16)] can be approximated into the following form,

$$\hat{H}_{\rm eff}/\hbar \approx \sum_{ij}\left[\mathcal{L}_{ij}\hat{a}^\dagger_i\hat{a}_j + \frac{1}{2}\mathcal{M}_{ij}\hat{a}^\dagger_i\hat{a}^\dagger_j + \frac{1}{2}\mathcal{M}^*_{ij}\hat{a}_i\hat{a}_j\right], \tag{S30}$$

where $i, j \in \{-5/2, 1/2, 3/2\}$, and $\hat{a}_-, \hat{a}^\dagger_-$ are not included because they only appear in the terms beyond the second order. Here, $\mathcal{L}$ is a Hermitian matrix, and $\mathcal{M}$ is a symmetric matrix. They take the following form:

$$\mathcal{L} = \begin{pmatrix} K_1 & K_4\sqrt{p_{5/2}} & 0\\ K_4\sqrt{p_{5/2}} & K_2 & K_5\sqrt{p_{-3/2}}\\ 0 & K_5\sqrt{p_{-3/2}} & K_3\end{pmatrix}, \quad \mathcal{M} = \begin{pmatrix} 0 & K_6\sqrt{p_{-3/2}} & K_6\\ K_6\sqrt{p_{-3/2}} & 2K_6\sqrt{p_{-3/2}p_{5/2}} & K_6\sqrt{p_{5/2}}\\ K_6 & K_6\sqrt{p_{5/2}} & 0\end{pmatrix}, \tag{S31}$$

where

$$\begin{gathered} K_1 = -\frac{16}{35}C_1 + \frac{16}{245}C_2, \quad K_2 = \frac{8}{35}C_1 + \left(\frac{144}{1225}p_{-3/2} + \frac{16}{245}p_{5/2}\right)C_2,\\ K_3 = K_5 = \frac{144}{1225}C_2, \quad K_4 = \frac{16}{245}C_2, \quad K_6 = \frac{48\sqrt{5}}{1225}C_2,\\ C_1 = \sum_{\nu=A,B}\frac{|\Omega_\nu|^2\Delta_\nu}{(\Delta_\nu - 13\chi N/35)^2}, \quad C_2 = \sum_{\nu=A,B}\frac{|\Omega_\nu|^2\Delta_\nu\chi N}{(\Delta_\nu - 13\chi N/35)^2[\Delta_\nu - (9+8p_{-3/2}\chi N/35)]}.\end{gathered} \tag{S32}$$

The dynamics of an operator $\hat{O}$ under Eq. (S30) can be solved exactly using the Heisenberg equations of motion $\partial_t\hat{O} = i[\hat{H},\hat{O}]/\hbar$, so we have the following set of equations for the bosonic creation and annilation operators,

$$\partial_t \begin{pmatrix} \hat{A} \\ \hat{A}^\dagger \end{pmatrix} = -i \begin{pmatrix} \mathcal{L} & \mathcal{M} \\ -\mathcal{M}^* & -\mathcal{L}^* \end{pmatrix} \begin{pmatrix} \hat{A} \\ \hat{A}^\dagger \end{pmatrix}, \tag{S33}$$

where $\hat{A} = (\hat{a}_{-5/2}, \hat{a}_{1/2}, \hat{a}_{3/2})^T$. This allows us to reduce the problem into the calculation of right eigenvalues and right eigenvectors for a non-Hermitian matrix,

$$\begin{pmatrix} \mathcal{L} & \mathcal{M} \\ -\mathcal{M}^* & -\mathcal{L}^* \end{pmatrix} \begin{pmatrix} \vec{u}_p \\ \vec{v}_p \end{pmatrix} = \epsilon_p \begin{pmatrix} \vec{u}_p \\ \vec{v}_p \end{pmatrix}, \tag{S34}$$

These are the so-called Bogoliubov-de Gennes (BdG) equations. By defining a matrix $\mathcal{T}$ whose column vectors are the right eigenvectors $(\vec{u}_p, \vec{v}_p)^T$, and a diagonal matrix $\mathcal{H}$ whose diagonal elements are the right eigenvalues ϵ_p, we have

$$\begin{pmatrix} \hat{A}(t) \\ \hat{A}^\dagger(t) \end{pmatrix} = \mathcal{U}_t \begin{pmatrix} \hat{A}(0) \\ \hat{A}^\dagger(0) \end{pmatrix}, \quad \mathcal{U}_t = \exp\left[-i \begin{pmatrix} \mathcal{L} & \mathcal{M} \\ -\mathcal{M}^* & -\mathcal{L}^* \end{pmatrix} t\right] = \mathcal{T} e^{-i\mathcal{H}t}\mathcal{T}^{-1}. \tag{S35}$$

Since we are considering the initial state as the vacuum state, the expectation values of any quadratic form of bosonic operators are given by

$$\left\langle \mathrm{vac} \middle| \begin{pmatrix} \hat{A}(t) \\ \hat{A}^\dagger(t) \end{pmatrix} \begin{pmatrix} \hat{A}(t) & \hat{A}^\dagger(t) \end{pmatrix} \middle| \mathrm{vac} \right\rangle = \mathcal{U}_t \begin{pmatrix} 0 & \mathcal{I} \\ 0 & 0 \end{pmatrix} \mathcal{U}_t^T, \tag{S36}$$

where $\mathcal{I}$ is the identity matrix. Using this method, we can calculate the short time dynamics of N_{diff}, which are shown with dashed lines in Fig. 3(c) in the main text.

Based on Eq. (S35), one can separate between the exponential growth (Phase I) and the sinusoidal oscillation (Phase II): If all the right eigenvalues ϵ_p are real numbers, the system is in Phase II; If at least one of the right eigenvalues ϵ_p are complex numbers, the system is in Phase I. According to Eq. (S34), if $(\vec{u}_p, \vec{v}_p)^T$ is a right eigenvector with right eigenvalue ϵ_p, $(\vec{v}_p^*, \vec{u}_p^*)^T$ is a right eigenvector with right eigenvalue $-\epsilon_p^*$. Since all the element of matrix $\mathcal{L}$ and $\mathcal{M}$ are real numbers, if ϵ_p is a right eigenvalue, ϵ_p^* is also a right eigenvalue. Therefore, if ϵ_p is a real number, we have a pair of right eigenvalues $\epsilon_p, -\epsilon_p$; if ϵ_p is a complex number, we have four right eigenvalues $\epsilon_p, -\epsilon_p, \epsilon_p^*, -\epsilon_p^*$. So we have two different cases for the six-level system discussed above: 2 real eigenvalues, 4 complex eigenvalues (Phase I); 6 real eigenvalues (Phase II). The phase boundary calculated by UPA generally agrees with the long-time averages $\overline{N}_{\mathrm{diff}} = \overline{N}_{3/2} - \overline{N}_{-5/2}$ and $\overline{N}_{\mathrm{sum}} = \overline{N}_{3/2} + \overline{N}_{-5/2}$ calculated by exact diagonalization with 20 atoms [see Fig. S3].

We can now explain the short-time behavior of chiral transport discussed in the main text based on the 4 pair creation processes described by matrix $\mathcal{M}$:

1) Process I: $\hat{a}^\dagger_{-5/2}\hat{a}^\dagger_{1/2}$ with strength $K_6\sqrt{p_{-3/2}}$

2) Process II: $\hat{a}^\dagger_{3/2}\hat{a}^\dagger_{1/2}$ with strength $K_6\sqrt{p_{5/2}}$

3) Process III: $\hat{a}^\dagger_{-5/2}\hat{a}^\dagger_{3/2}$ with strength K_6

4) Process IV: $\hat{a}^\dagger_{1/2}\hat{a}^\dagger_{1/2}$ with strength $K_6\sqrt{p_{-3/2}p_{5/2}}$

Notice that only process I and II change the value of N_{diff}, and the relative strength between these two processes can be tuned by $p_{-3/2}$, while Δ_B is another control knob to determine whether these processes are on resonance or not. When these two processes are both on resonance, e.g. $\Delta_B/\chi N \in (3, 4)$, the direction of chiral transport is fully tunable via $p_{-3/2}$ [see Fig. S3(a)]. When process I is off resonance, e.g. $\Delta_B/\chi N \in (5, 6)$, we always find chiral transport to the right side in the exponential growth regime, and balanced transport is only possible to achieved by tuning the whole system to sinusoidal oscillation regime [see Fig. S3(a)].

S3. NUMERICAL RESULTS FOR DYNAMICAL PHASE TRANSITION

Here we present additional numerical results using exact diagonalization that complements the discussion of the dynamical phase transition (DPT) discussed in Fig. 2 of the main text. Since exact diagonalization is only possible

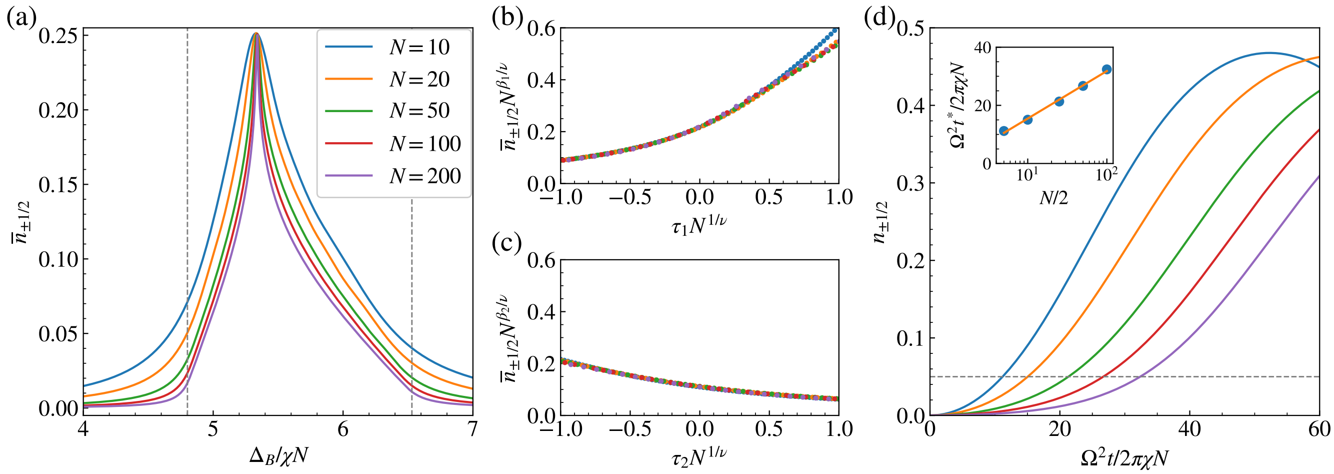

FIG. S4. Finite size scaling of DPT by varying atom number N while keeping χN fixed. (a) The long-time average fractional atom population of $|g_{\pm 1/2}\rangle$ states ($\overline{n}_{\pm 1/2}$) at $\Delta_A = -4\chi N$. The dashed lines indicate the dynamical critical points predicted by UPA in the previous section. (b,c) Data collapse of finite-size calculations of $\overline{n}_{\pm 1/2}$ near the critical points $\Delta_{B,1}$ and $\Delta_{B,2}$ (see text). (d) Short-time dynamics of $n_{\pm 1/2}$ at $\Delta_A = -4\chi N$ and $\Delta_B = 5.3\chi N$. The dashed line marks $n_{\pm 1/2} = 0.05$, which determines the typical time scale t^* for the obsevation of the DPT. The inset shows the logarithmic scaling of t^*.

for a small atom number, we explore the properties of this DPT in the thermodynamic limit using finite size scaling. Given the collective nature of the cavity-mediated interactions, it is convenient to vary the atom number N while keeping χN fixed. As we described in the main text, we consider the initial state $|g_{-3/2}\rangle^{\otimes(N/2)}|g_{3/2}\rangle^{\otimes(N/2)}$, which is an eigenstate of this system when $\Delta_A, \Delta_B \to \infty$. We choose $\Omega_A = \Omega_B = \Omega = 0.05\chi N$, and then perform a sudden quench to $\Delta_A = -4\chi N$ and Δ_B between $4\chi N$ and $7\chi N$. The order parameter of this DPT is the long-time average of the fractional atom population of the $|g_{\pm 1/2}\rangle$ states,

$$\overline{n}_{\pm 1/2} = \lim_{T\to\infty} \frac{1}{T}\int_0^T \frac{N_{\pm 1/2}(t)}{N} dt, \tag{S37}$$

shows a sharp change behavior as we varying Δ_B [see Fig. S4(a)]. Near the dynamical critical points ($\Delta_{B,1} = 4.80\chi N$, $\Delta_{B,2} = 6.53\chi N$), it is convenient to assume $\overline{n}_{\pm 1/2}$ as a homogeneous function,

$$\overline{n}_{\pm 1/2} = N^{-\beta_{1,2}/\nu} f(\tau_{1,2} N^{1/\nu}), \tag{S38}$$

where $\tau_1 = (\Delta_B - \Delta_{B,1})/\chi N$, $\tau_2 = (\Delta_B - \Delta_{B,2})/\chi N$. The data collapse of finite-size calculations are shown in Fig. S4(b,c), which gives $\beta_1 = 1.15$, $\beta_2 = 1.04$, $\nu = 2.38$. Based on the data collapse, one can conclude that

$$\begin{aligned} &\overline{n}_{\pm 1/2} = 0 \ (\tau_1 < 0), \quad \overline{n}_{\pm 1/2} \propto \tau_1^{\beta_1} \ (\tau_1 > 0), \\ &\overline{n}_{\pm 1/2} \propto (-\tau_2)^{\beta_2} \ (\tau_2 < 0), \quad \overline{n}_{\pm 1/2} = 0 \ (\tau_2 > 0), \end{aligned} \tag{S39}$$

at thermodynamic limit with $\tau_{1,2} \to 0$.

In experiments, it is easier to explore the DPT using short-time dynamics instead of long-time averages. In phase I, the exponential growth of $n_{\pm 1/2}$ is triggered by quantum fluctuations in the initial state, and therefore there is a delay time for $|g_{\pm 1/2}\rangle$ states to accumulate a macroscopic population [see Fig. S4(d)]. The delay time increases as we increase the atom number N. A reasonable measure of the delay time would be the time t^* for $n_{\pm 1/2}$ reaching 0.05 ($\sim 10\%$ of maximum value). As shown in the inset of Fig. S4(d), we find $t^* \propto \ln(N/2)$.

S4. EXPERIMENTAL CONSIDERATIONS

A. Experimental parameters

Here we discuss the specific case of the $^1S_0 \to {}^3P_1$ transition in ^{87}Sr using the experimental parameters described in Ref. [S2]. We show that current experimental parameters lie in the regime where our theory predictions are valid and the dynamics is dominated by unitary evolution. The conditions are summarized in Table S1. In this system the typical atom-light coupling strength is $g_c \sim 2\pi \times 10$kHz, and the cavity intensity decay rate is $\kappa \sim 2\pi \times 100$kHz. Choosing atom number $N \sim 2 \times 10^5$, and cavity detuning $\Delta_c = 2\pi \times 20$MHz, we ensure the adiabatic elimination of the cavity photons and dominant unitary dynamics compared to cavity decay. As we discussed in the main text, setting $\Omega_{A,B} = \Omega = 0.05\chi N$, and $\Delta_{A,B}$ comparable or larger than χN satisfy the requirement of adiabatic elimination of atomic excited states. Note that the correlated hopping processes occur at a rate $\Omega^2\chi N/\Delta_{A,B}^2$, while the dissipation process in atomic ground manifold due to spontaneous emission from excited manifold occur at a rate $\Omega^2\gamma/\Delta_{A,B}^2$. Since for the above parameters $\chi N \sim 2\pi \times 1$MHz, is more than two orders of magnitude larger than the spontaneous emission rate $\gamma = 2\pi \times 7.5$kHz of $^3P_1$ states, we can ignore dissipation during the time scale of interests. As shown in Fig. 4 in the main text, the correlation spreading takes place at a time scale $\Omega^2 t/2\pi\chi N \sim 100$ for atom number $N = 10$, and the corresponding delay time is $\Omega^2 t^*/2\pi\chi N \sim 10$. For $N \sim 2 \times 10^5$, the delay time is expected to be 7 times longer, and the relevant experimental time scale to observe correlation spreading is $\Omega^2 t/2\pi\chi N \sim 200$, which corresponds to $t \sim 80$ms.

TABLE S1. Summary of approximations in theory model and the required parameter regimes (see text for details)

Theory approximations	Required parameter regimes
Adiabatic elimination of cavity photons	$\lvert\Delta_c\rvert \gg g_c\sqrt{N}$
Negligible cavity loss	$\lvert\Delta_c\rvert \gg \kappa$
Adiabatic elimination of atomic excited states	$\lvert\Delta_{A,B}\rvert, \lvert\chi N\rvert \gg \lvert\Omega_{A,B}\rvert$
Negligible spontaneous emission	$\lvert\chi N\rvert \gg \gamma$
Weak magnetic field limit	$\lvert\chi N\rvert \gg \delta_e F_e$

Although we have assumed a vanishing magnetic field in our discussions, our protocol is relatively robust against the magnetic field along the quantization axis ($\hat{z}$), because the effective ground state Hamiltonian [Eq. (S16)] commutes with linear Zeeman shifts on the ground manifold. The only constraint for the maximum tolerable magnetic field is the Zeeman shifts of the excited manifold. They should be small compared to χN, otherwise the many-body structure of the atomic excited states will change and the Zeeman splitting must be included in the derivation of the effective Hamiltonian. We benchmark the effect of magnetic field using the following Hamiltonian,

$$\hat{H} = \hat{H}_{\text{aa+d}} + \hat{H}_Z, \quad \hat{H}_Z/\hbar = \delta_e \hat{F}_e^z + \delta_g \hat{F}_g^z, \tag{S40}$$

where $\hat{H}_{\text{aa+d}}$ is defined in Eq. (S3), $\delta_e = \mathcal{G}_{F=9/2,{}^3P_1}\mu_B B$, and $\delta_g = \mathcal{G}_{F=9/2,{}^1S_0}\mu_B B$ with μ_B the Bohr magneton, and B the magnetic field. Here, $\mathcal{G}_{F=9/2,{}^3P_1} = 2/33$, $\mathcal{G}_{F=9/2,{}^1S_0} = -1.3 \times 10^{-4}$ are the Landé g-factors for ^{87}Sr atoms [S3]. We have calculated the exact short-time dynamics for 4 atoms using Trotterization, which are shown as solid lines in Fig. S5. For the case of 0.1G magnetic field [see Fig. S5(a)], the Hamiltonian dynamics agrees with the case of zero magnetic field at short time. Even with a 0.6G magnetic field [see Fig. S5(b)], it is still possible to observe similar dynamics generated by correlated hopping processes. A larger magnetic field can be problematic since it leads to a significant suppression of the correlated hopping processes [see Fig. S5(c)].

B. Initial state preparation

Here we discuss the experimental protocol for initial state preparation. The most common type of initial state required in the main text is half of atoms in $m_F = -F$ state and half of atoms in $m_F = F$ state. Here we use ^{87}Sr atom ($F = 9/2$) as an example to explain how to prepare it via optical pumping during the laser cooling procedure. Such type of initial state preparation has already been demonstrated in previous experiments [S4, S5]. Note that $^1S_0 \to {}^3P_1$ transition is a common choice for narrow-line cooling. If we use π-polarized light to interrogate the transition $^1S_0(F = 9/2) \to {}^3P_1(F' = 7/2)$, one can see that the $m_F = \pm 9/2$ states in the ground manifold are the dark states during the cooling process, and atoms will spontaneous decay and accumulate there at the end of cooling process. So one can prepare roughly $N/2$ atoms in $m_F = -9/2$ state, and $N/2$ atoms in $m_F = 9/2$ state due to the symmetric Clebsch-Gordan coefficients. Note that this process generates no coherence between $m_F = \pm 9/2$ states due to dissipation, which allows us to generate the state $|g_{-9/2}\rangle^{\otimes(N/2)}|g_{9/2}\rangle^{\otimes(N/2)}$, up to an statistical atom number imbalance of the order of $\sqrt{N}$. This small imbalance in the initial state has negligible effects on the physical phenomena we are investigating.

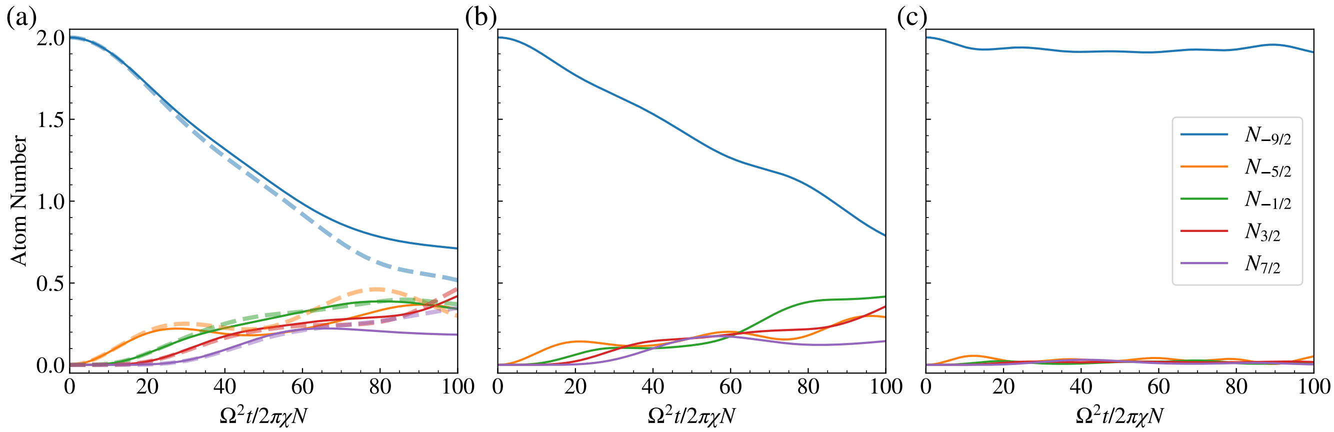

FIG. S5. Effect of a finite magnetic field in the short-time dynamics for the case of ^{87}Sr atoms. The initial state is $|g_{-9/2}\rangle^{\otimes(N/2)}|g_{9/2}\rangle^{\otimes(N/2)}$, and $N=4$. We set $\chi N = 2\pi \times 1$MHz, $\Delta_A = -3\chi N$, $\Delta_B = 4.1\chi N$. The solid lines are calculated based on Eq. (S40), with magnetic field (a) $B = 0.12$G ($\delta_e = 0.01\chi N$), (b) $B = 0.59$G ($\delta_e = 0.05\chi N$), and (c) $B = 1.77$G ($\delta_e = 0.15\chi N$). The dashed lines in (a) are the ideal case with zero magnetic field.

For the general initial states with half of atoms atoms on the upper leg and half of atoms atoms on the lower leg, $|\psi\rangle = |\psi_{\rm up}\rangle \otimes |\psi_{\rm down}\rangle$, one can first prepare half of atoms in $m_F = -F$ state and half of atoms in $m_F = F$ state as we discussed above, and then use the ultranarrow $^1S_0 \rightarrow {}^3P_0$ transition to perform coherent transfer. This ultranarrow transition allows us to energetically resolve each of the hyperfine levels even with a small magnetic field.

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